Exploratory Data Analysis of Taylor Swift's Spotify Catalog

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1 Introduction and Dataset Description

1.1 Rationale, Applications, and Collection Background

Digital streaming platforms have revolutionized how musical trends and artistic performance are quantified, empowering creators and record labels to make data-driven decisions that mitigate commercial risks. This research conducts an Exploratory Data Analysis (EDA) on Taylor Swift's Spotify catalog, extracted from Kaggle via the Spotify Web API using the `jarredpriester/taylor_swift-spotify-dataset` repository. Her multi-decade career and diverse musical style make her catalog an ideal empirical landscape to explore how acoustic features interact with audience engagement over time.

1.2 Variable Specification and Typologies

The dataset comprises 582 rows and 18 columns. The variables are classified into four metadata-defined categories:

Primary Metric of Interest (`int64`): The variable `popularity` (0–100) represents a dynamic score calculated by Spotify based on stream volume and recency. It serves as the exclusive proxy for audience engagement and commercial impact within this study.

Audio Descriptors (`float64`): Continuous acoustic features engineered by Spotify: `acousticness`, `danceability`, `energy`, `instrumentalness`, `liveness`, `speechiness`, and `valence` (all normalized from 0.0 to 1.0), alongside physical descriptors `loudness` (negative decibels, dB) and `tempo` (beats per minute, BPM).

Structural Layout Parameters (`int64`): The variables `track_number` (album sequence position) and `duration_ms` (track length in milliseconds).

Identification and Metadata Parameters (`object`): Nominal text strings including `name`, `album`, `id`, `uri`, and `release_date`, as well as `Unnamed: 0` (`int64` system row identifier with no behavioral value).

1.3 Challenges and Data Quality Limitations

The raw dataset contains zero missing values and zero duplicate rows. However, two major data quality concerns were identified:

Non-Musical Data Inconsistencies: Initial scans revealed a small subset of records carrying a popularity score of exactly 0, which qualitative checks confirmed represent 3 non-musical voice memos from the *1989 (Deluxe Edition)* album ("the outliers").

Redundancy and Representation Bias: The API captures multiple published editions of the same album, causing identical track entries to appear repeatedly due to multi-version catalog re-recordings, creating severe chronological over-representation bias.

2 Data Cleaning and Preprocessing

2.1 Outlier Detection and Invalid Data Isolation

To ensure the integrity of the catalog, a systematic approach for outlier detection was established. This phase focuses on identifying data quality limitations, which consist of the previously defined outliers. Rather than removing these outliers during this exploratory phase, they are retained to assess their impact on the dataset's overall variance.

2.2 Temporal Feature Extraction and Type Conversion

The raw `release_date` column was loaded as a generic text string. To unlock its time-series potential, a multi-step temporal engineering pipeline was implemented. First, a slicing operation extracted the first four characters, which were cast as an integer to construct a new continuous column named `release_year`. This conversion was necessary because raw text strings are mathematically inert and cannot be used to calculate statistical trends or compute correlations. Transforming this dormant text into an active numerical element is the cornerstone of this study, enabling the evaluation of macro-level trends and numerical correlations with popularity. Second, the complete raw string was converted into a standardized continuous pandas Datetime object (`release_date_D`) to preserve precise day and month metadata. This functional separation ensures that while `release_year` is optimized for macro-level statistical computations, `release_date_D` allows the construction of high-resolution scatter plots with precise, unflattened sub-yearly visual placement.

2.3 Redundant Resolution and Label Optimization via Structural Album Clustering

The raw dataset contains several different versions of the same album due to multi-version catalog re-recordings, causing identical compositions to appear multiple times. To resolve this categorical fragmentation and optimize visualization space, a specialized string-matching pipeline consolidated these variations into 12 high-level categorical groups under a new column named `album_families`. Under this framework, the three previously identified outliers are sustained within the consolidated *1989* album family. Furthermore, over-length titles were programmatically shortened to prevent visual overlapping in plots (specifically compressing THE TORTURED POETS DEPARTMENT to *TTPD*, Taylor Swift (Deluxe Edition) to *Deluxe*, and Live From Clear Channel to *Live*).

Analyzing raw album names directly would introduce severe data fragmentation and extreme visual clutter, splitting cohesive musical eras into superficial sub-pockets and exhausting critical page layout margins. Establishing these 12 clean, compact groups resolves both fragmentation and spatial formatting limitations, allowing downstream multivariate box plots to remain clean and readable.

3 Univariate Analysis

3.1 Distributional Architecture of Track Popularity

To evaluate the central tendencies and shape of public engagement, the continuous variable `popularity` was visualized (Figure 1).

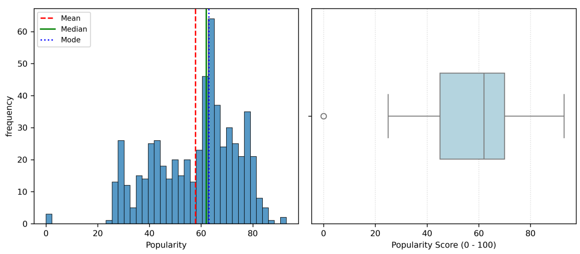

Figure 1: Univariate distribution of track popularity combining (a) histogram and (b) horizontal box plot.

The descriptive statistical summary reveals a powerful upper-register density with a Mean of 57.86, Median of 62.00, Mode of 63.00, Minimum of 0.00, Maximum of 93.00, Standard Deviation of 16.15, and Skewness of -0.533. As illustrated in the histogram in Figure 1(a), the distribution exhibits a distinct left-skewed shape, driven by the outliers. While the histogram suggests potential contemporary outliers on the far-right tail (above 90), the box plot in Figure 1(b) mathematically contextualizes this behavior. Due to the catalog's high streaming success, the calculated upper statistical fence stands at a theoretical 107.5, which lies beyond the physical 100-point ceiling of Spotify's metric. Consequently, extremely successful tracks on the right-hand tail are not mathematical outliers, leaving the three voice memos below the lower fence of 7.5 as the only true global anomalies in the raw dataset. Their presence artificially inflates the dataset's variance and standard deviation, demonstrating the critical necessity to implement a programmatic filter before subsequent statistical modeling to ensure a robust predictive analysis.

3.2 Distributional Architecture of Danceability

To evaluate the shape of the continuous variable `danceability`, the distribution was visualized (Figure 2).

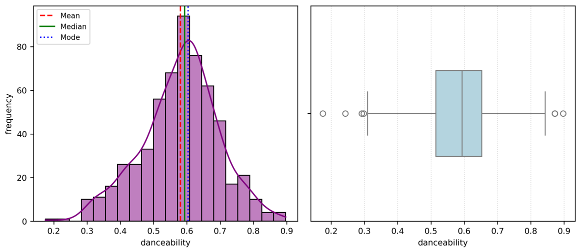

Figure 2: Univariate distribution of track danceability combining (a) histogram and (b) horizontal box plot.

The statistical summary reveals a Mean of 0.581, Median of 0.594, Mode of 0.604, Minimum of 0.175, Maximum of 0.897, Standard Deviation of 0.115, and Skewness of -0.265, indicating an approximately symmetric, balanced bell-curve distribution around the 0.58 to 0.60 region. However, the box plot in Figure 2(b) isolates statistical outliers on both ends, defined by a lower fence of 0.3080 and an upper fence of 0.8600. The upper-fence outliers reveal a critical redundancy: a single track ("Vigilante Shit", 0.8726) appears three times in this upper outlier pool across different expanded editions of the *Midnights* family. This repeated occurrence provides clear empirical evidence of representation bias in the raw data, directly justifying the need for the categorical family consolidation and data cleaning performed in Section 2.3.

3.3 Distributional Architecture of Album Families

To evaluate the sample distribution across categorical groupings, the univariate frequency of the engineered variable `album_families` was visualized using a sorted horizontal count plot (Figure 3).

The tracks are distributed across 12 unique `album_families`, with the *1989* family serving as the statistical mode, appearing 75 times (72 musical tracks plus the outliers). As illustrated

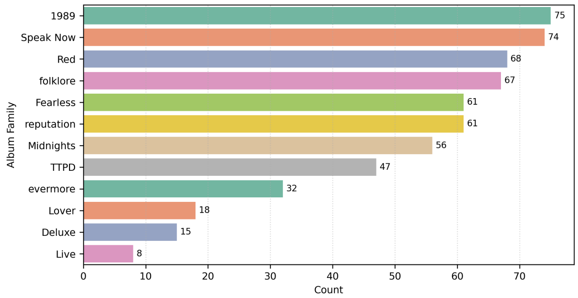

Figure 3: Univariate frequency distribution of tracks across engineered album families.

in Figure 3, the count plot exposes a severe sample imbalance within the dataset, dominated by a few highly populated families: *1989* (75 tracks), *Speak Now* (74 tracks), *Red* (68 tracks), and *Folklore* (67 tracks). Identifying this distributional variance is computationally critical; it alerts that subsequent statistical comparisons of popularity will have highly unequal group representations, meaning that any calculations computed for smaller families (like *Live* with only 8 tracks) will inherently carry higher statistical variance, a limitation that must be accounted for during predictive modeling.

4 Multivariate Analysis

4.1 Numeric vs. Numeric Relationship: Correlation Analysis

To explore continuous feature interactions and evaluate linear dependencies, a Pearson correlation matrix was computed and visualized via a heatmap (Figure 4).

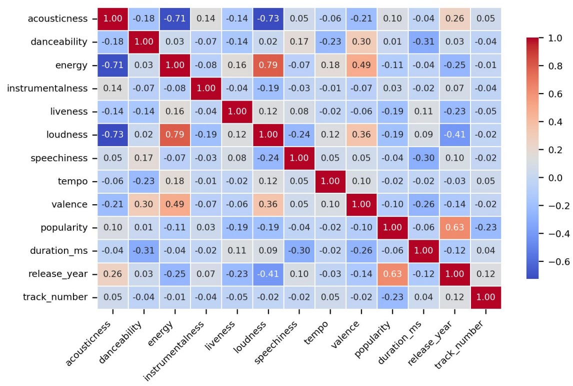

Figure 4: Pearson correlation matrix heatmap for continuous audio characteristics and popularity.

The objective of this screening is to detect patterns of association and identify potential multicollinearity that could violate statistical assumptions in predictive models. The matrix reveals three prominent relationships:

Energy and Loudness: A strong positive Pearson correlation coefficient of 0.79 is observed, signaling a notable risk of multicollinearity between these two features that must be monitored during feature selection.

Loudness and Acousticness: A strong negative relationship is identified, with a Pearson correlation coefficient of -0.73 (coupled with a -0.71 correlation between `energy` and `acousticness`), showing that higher acoustic values correspond to lower loudness and energy.

Popularity and Release Year: The primary metric of interest, `popularity`, displays its strongest linear association with `release_year`, yielding a positive Pearson correlation of 0.63.

In contrast, all other continuous acoustic variables (such as `duration_ms`, `track_number`, `danceability`, and `tempo`) display near-zero or weak linear correlations with popularity (ranging between -0.23 and 0.10). This indicates that a song's baseline popularity is heavily associated with its release timeframe rather than its specific standalone audio features, a critical insight for subsequent predictive modeling tasks.

4.2 Numeric vs. Numeric Relationship: Popularity and Release Temporal Dynamics

To visually unpack the strong correlation ($r = 0.63$) identified in the correlation matrix, Figure 5 maps individual track popularity against its release date, categorized by `album_families`. The collective release of tracks within specific album launch years naturally forms distinct vertical clusters on the scatter plot, allowing for a structured temporal evaluation of the catalog.

The visualization exposes two highly critical dynamics regarding how track popularity is distributed over time, alongside serving as a diagnostic tool to isolate structural anomalies:

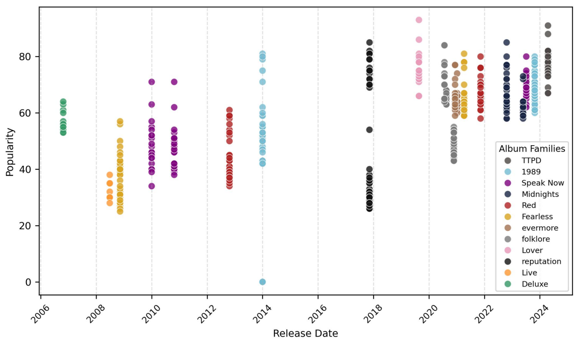

Figure 5: Scatter plot mapping track popularity against release date, categorized by album families.

The Recency Effect: A clear upward progression is visible from left to right, where tracks released in earlier eras (2006–2012) exhibit lower and more widely dispersed popularity scores, primarily clustering below the 60 threshold. Conversely, tracks released from 2020 onward exhibit significantly tighter, elevated distributions, with the majority of songs sustaining elevated popularity scores heavily clustering between approximately 60 and 80. This temporal shift visually validates the positive correlation coefficient, indicating that contemporary releases and recent catalog additions heavily drive active listener engagement.

The Re-recording Album Impact: The projection of `album_families` reveals a prominent structural phenomenon. For families with re-recorded releases, most notably *1989* and *Red*, dual vertical columns are observed. Comparing these columns reveals that while the original tracks from 2014 and 2012 are widely dispersed and drop heavily below the 60 threshold, their modern re-recorded counterparts (released in 2023 and 2021) experience a significant upward shift, remaining tightly elevated and heavily clustered above 60.

The Outlier/Voice Memo Impact: The plot successfully isolates the outliers.

Collectively, these visual dynamics provide robust evidence that popularity is highly dependent on both temporal recency and strategic catalog re-releases, which serves as a vital feature-engineering insight for subsequent predictive modeling.

4.3 Categorical vs. Numeric Relationship: Popularity Across Album Families

To analyze the categorical influence on the continuous primary metric of interest, Figure 6 displays the distribution of track popularity across specific `album_families`.

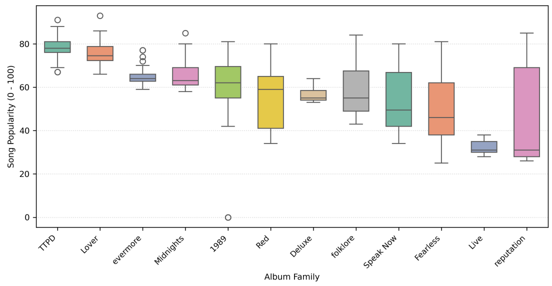

Figure 6: Box plot distribution of track popularity across categorized album families.

Distributional Variance and Group Differences: The box plot reveals clear localized variance across album families. Contemporary families like `TTPD` (2024) and `Lover` (2019) display compressed, highly elevated boxes, indicating consistent streaming. Conversely, the `reputation` family (2017) exhibits the largest variance in the catalog, spanning scores from 20 to the mid 80s, showcasing highly polarized audience engagement despite its single release window.

The Impact of Re-recordings: Specific families, most notably *Red* and *Speak Now*, combine tracks from original releases (2012 and 2010) alongside their modern re-recordings (2021 and 2023). The re-recorded tracks maintain substantially higher popularity scores than their original counterparts. Consequently, combining them into a single box creates a highly spread-out distribution with wider interquartile ranges (IQR) spanning from low to high popularity scores, explaining the prominent variance observed in these specific families.

Anomalies Within Specific Album Families: Evaluating specific localized boundaries exposes extreme performance polarization within single thematic concepts, especially when contrasted against the overall global popularity scale (mean of 57.86 and a median of 62.00):

The *evermore* Family Anomaly (2020): Localized screening yields a lower fence of 57.88 and an upper fence of 70.88. While the distribution presents no anomalies below the lower boundary, the track "willow" (popularity: 77) bypasses the upper threshold. Although this score is standard on the global scale, it represents a highly significant positive anomaly within the context of the quieter, indie-folk-leaning *evermore* family.

The *TTPD* Family Anomaly (2024): Localized screening yields a lower fence of 68.50 and an upper fence of 88.50. This method successfully isolates anomalies on both ends of the spectrum. The track "Fortnight (feat. Post Malone)" (popularity: 91) sits above the upper fence, while "Clara Bow" (popularity: 67) sits below the lower fence. Although 67 is well above the global scale, it represents an underperforming deep-cut anomaly within the highly streamed, contemporary environment of this specific era.

Statistical Modeling Implications: Under the flat, global screening applied in Figure 1, the only isolated anomalies were the outliers. A global screening completely masks unique, localized anomalies within specific eras.

Ultimately, this yields a dual statistical insight: while newer releases generally sustain higher popularity, temporal progression alone cannot explain catalog dynamics. Popularity is heavily governed by localized era characteristics, proving a micro level framework is a necessity. Modeling these subgroups prevents pooling bias, allowing downstream machine learning models to distinguish general stylistic baselines from true anomalies.

5 Conclusion and Insights

This EDA successfully established a robust operational baseline for analyzing structural, temporal, and commercial dynamics within Taylor Swift's Spotify catalog. In the initial univariate stage, evaluating the continuous popularity metric through Figure 1 illustrated two distinct statistical views. The histogram in Figure 1(a) revealed a negatively skewed distribution characterized by a skewness of -0.533, suggesting potential upper-register outliers, but the global box plot in Figure 1(b) mathematically disproved this by establishing an upper statistical fence of 107.5, well beyond the physical 100-point limit of Spotify. Consequently, the box plot isolated the outliers directly addressing the non-musical data inconsistency limitation. To address the second major data challenge of redundancy and representation bias, which was caused by identical tracks appearing repeatedly across catalog re-recordings, I engineered the `album_families` feature. The necessity of this engineering was first highlighted in the univariate analysis of `danceability` in Figure 2. While the histogram in Figure 2(a) revealed an approximately symmetric, balanced bell-curve distribution characterized by a skewness of -0.265, the box plot in Figure 2(b) isolated statistical outliers on both ends of the spectrum, exposing severe chronological redundancy through the repeated appearance of the track "Vigilante Shit" in the upper outlier pool. Grouping the catalog into twelve engineered `album_families` in Figure 3 effectively resolved fragmentation (*1989* mode: 75, *Live* min: 8). While eliminating duplicate redundancy, this consolidation exposes sample imbalance and higher IQR variance in smaller families ($n = 8$), requiring minimum category sample thresholds in future modeling.

To unlock the dataset's time series potential, I implemented a temporal engineering pipeline during the preprocessing stage to extract the numeric `release_year` from the raw text, alongside converting the full string into a standardized datetime object. This active numerical element enabled the subsequent multivariate phase, where the correlation heatmap in Figure 4 identified a strong linear association between `popularity` and `release_year` with a coefficient of 0.63, while also warning of a notable multicollinearity risk between `energy` and `loudness` with a high correlation of 0.79. To unpack these temporal dynamics, mapping individual track popularity against release dates in Figure 5 visually confirmed the recency effect, showing that contemporary releases sustain tighter, more elevated popularity scores. Crucially, Figure 5 also exposed the dramatic commercial impact of strategic catalog re-releases, where I observed that the re-recorded tracks within these specific families clustered at much higher popularity ranges than their original counterparts, illustrating how temporal shifts and strategic decisions interact to shape catalog performance.

Transitioning from the global pooling of Figure 1 to the group based benchmarking of Figure 6 mathematically proved that each respective era possesses a fundamentally unique mean, variance, and internal dispersion pattern, forming distinct trends per album family. Localized cross era profiling provides the definitive statistical justification for rejecting aggregate baselines. As demonstrated by the empirical boundary profiles of the *evermore* and *TTPD* families, aggregate metrics naturally obscure the presence of localized anomalies, such as individual viral hits rising above expected thresholds or deep cuts suffering structural popularity penalties within highly performing eras. Ultimately, Figure 6 stands as the most informative diagnostic climax of this study, demonstrating that understanding local distributions is essential to capture the true catalog behavior.

Consequently, these EDA findings directly inform potential downstream predictive modeling tasks by outlining critical data structuring requirements. Evaluating the variables at a localized micro level, rather than through global macro aggregates, is a statistical necessity; incorporating the `album_families` feature is crucial to prevent aggregate pooling bias. Furthermore, acknowledging the high statistical variance in smaller families (e.g., *Live*, $n = 8$), future predictive tasks should establish minimum sample thresholds per album family to mitigate sample imbalance noise. To ensure the stability of linear models, the outliers must be removed to prevent severe parameter distortion. Furthermore, the high correlation between `energy` and `loudness` requires strategic handling. While this collinearity poses a multicollinearity risk in linear regression that may necessitate feature reduction or transformation, both features can be safely retained in tree-based ensemble models.Ultimately, downstream predictive tasks must focus on modeling track popularity as the core target variable, recognizing that temporal progression outweighs individual standalone acoustic features as a primary driver (Figure 4), whose predictive power is best captured when contextualized dynamically within individual album families.